

Problem A:How to win the ‘2048’?

The intelligence game ‘2048’ is sweeping the global. Plenty of players claim that they cannot help beginning another attempt, getting 2048! 2048 is played on a simple gray 4×4 grid with simple principles: all tiles slide to the same direction at each move; if two tiles of the same number collide while moving, they will merge into a tile with the total value of the two tiles that collided; and a new tile will randomly appear in an empty spot on the outer frame of the board with a value of either 2 or 4 after each turn. The game is won when a tile with a value of 2048 appears on the board. On the contrary, if there are no empty spaces and no adjacent tiles with the same value (i.e. no more moves are allowed), the game ends.

Build a mathematical model to deal with the following questions:

1. Provide a model in order to win this game. To test the performance of your algorithm, you should consider the number of moves and the probability of winning this game.
2. If further moves are allowed when a value of 2048 appears, what will be the maximum value of a tile? If the game’s grip is extended to $N \times N$, what will be the maximum value of a tile?